Lecture 05, Math 441

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Substitutions Methods (cont.)

$y' = F(\frac{y}{x})$ Try $v(x) = \frac{y(x)}{x} \rightarrow$ separate

Consider $x\,y' = \frac{y^2}{x} + y$ and rewrite as $y' = \frac{y^2}{x^2} + \frac{y}{x} \rightarrow y' - \frac{1}{x}y = \frac{1}{x^2}y^2$

This is an example of a Bernoulli equation. Use the substitution

$v(x) = y^{1-2} = \frac{1}{y}$:

$y = \frac{1}{v}$; $\quad y' = -\frac{1}{v^2}v'$ and substituting in the equation:

$$-\frac{1}{v^2}v' - \frac{1}{xv} = \frac{1}{x^2v^2}; \rightarrow \; v' + \frac{1}{x}v = -\frac{1}{x^2}$$

which is linear and can be solved with an integrating factor. timestamp: 00:00

The integrating factor is $\mu(x) = \mathrm{e}^{\int \frac{1}{x}\mathrm{d}x} = \mathrm{e}^{\ln(x)} = x$

Multiplying the equation for v by μ we get:

$$x\,v' + v = -\frac{1}{x} \;\rightarrow\; (x\,v)' = -\frac{1}{x} \;\rightarrow\; x\,v = -\ln(x) + A \;\rightarrow$$

$$v(x) = \frac{A - \ln(x)}{x} \;\rightarrow\; y(x) = \frac{x}{A - \ln(x)}$$

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Bernoulli Equations

In general, a Bernoulli equation has the form

$$y' + p(x)\, y = q(x)\, y^{\alpha}$$

where $\alpha \neq 0, 1$ (If α is 0 or 1, this would be a linear equation)

We can use the substitution $v(x) = y^{1-\alpha}$ and the equation becomes

1st order linear for $v(x)$ and we can use an integrating factor.

Ex: $y' = \frac{y}{x} + \sqrt{y} \;\rightarrow\; y' - \frac{1}{x}y = y^{\frac{1}{2}}$

So $\alpha = \frac{1}{2}$ and we use $v(x) = y^{1-\frac{1}{2}} = y^{\frac{1}{2}}$; $y = v^2$; $y' = 2v\,v'$

Using this substitution to get

$$2v\,v' - \frac{1}{x}v^2 = v \;\rightarrow\; v' - \frac{1}{2x}v = \frac{1}{2}$$

(assuming $v \neq 0$. If $v = 0$ then $y = 0$ and you can verify this is another solution)

The integrating factor is $\mu(x) = \mathrm{e}^{-\int \frac{1}{2x}\mathrm{d}x} = \mathrm{e}^{-\frac{1}{2}\ln(x)} = x^{-\frac{1}{2}} = \frac{1}{\sqrt{x}}$ and

use this to get:

$$\frac{v'}{\sqrt{x}} - \frac{1}{2x\sqrt{x}}v = \frac{1}{2\sqrt{x}} \;\rightarrow\; \left(\frac{1}{\sqrt{x}}v\right)' = \frac{1}{2\sqrt{x}} \;\rightarrow\; \frac{1}{\sqrt{x}}v = \sqrt{x} + A$$

$$v = x + A\sqrt{x} = \sqrt{y} \;\rightarrow\; \boxed{y(x) = \left(x + A\sqrt{x}\right)^2}$$

to which we need to add the singular solution $\boxed{y(x) = 0}$.

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Theorem of Existence and Uniqueness for 1st Order Linear IVP

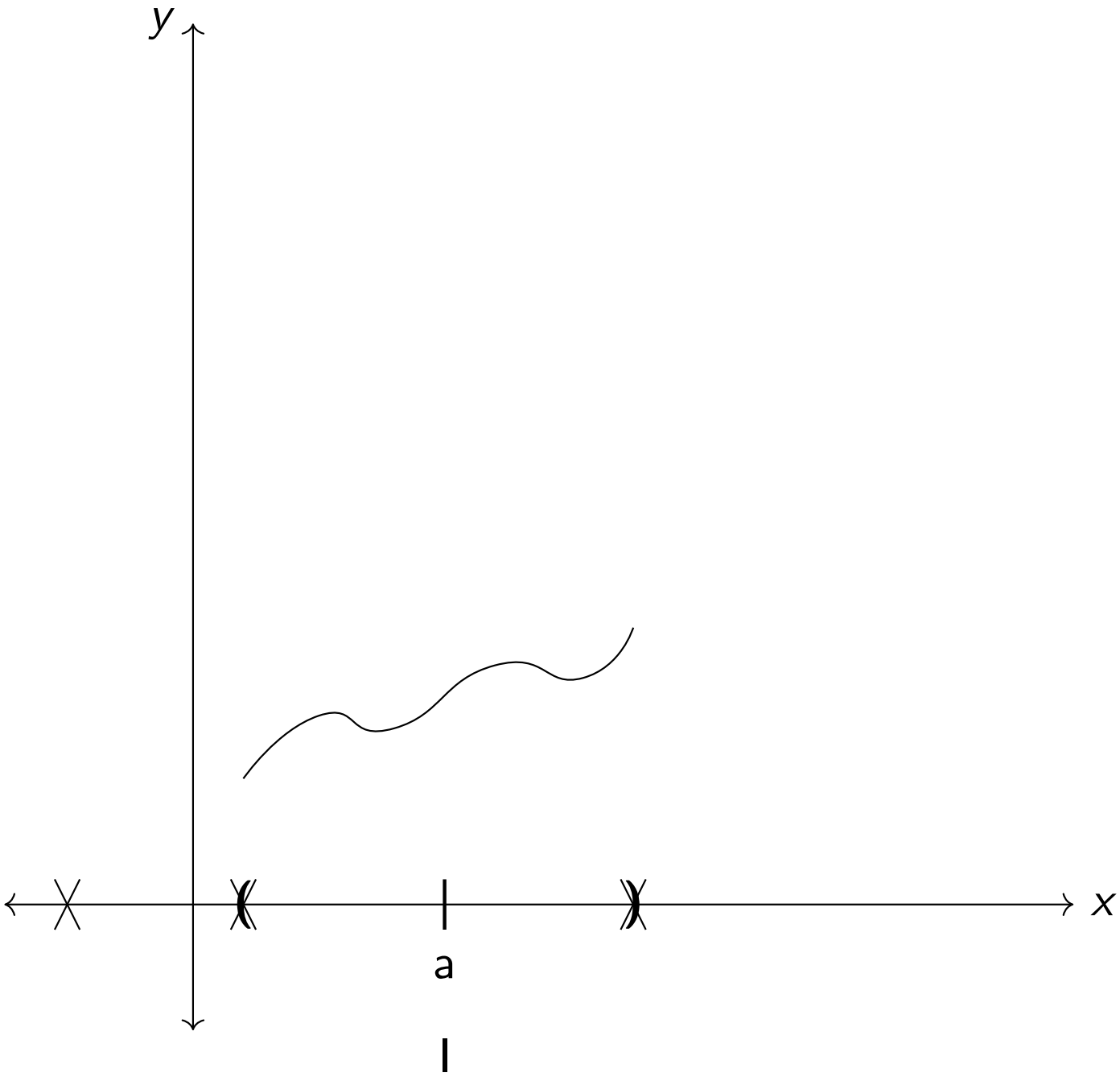
Consider the 1st order linear initial value problem

$$\begin{cases} y' + p(x)y = q(x) \\ \\ y(a) = b \end{cases}$$

Theorem: If $\exists$ an interval I around $x = a$ where p and q are continuous, then $\exists$ one and only one solution to the IVP over all of I. ($\exists$ means "it exists" or "there is")

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Note: the interval I can be determined without solving the equation, it depends only on the interval of continuity of p and q. So we can a priori know where the solution is guaranteed to exist and be unique.



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We can prove this by just finding the solution of the IVP using the method of integrating factor.

$\mu(x) = \mathrm{e}^{\int p \mathrm{d}x}$

$$\mu y' + p\mu y = \mu q \;\rightarrow\; (\mu y)' = \mu q \;\rightarrow\; \mu y = \int \mu q \mathrm{d}s + A$$

$$\boxed{y(x) = \frac{1}{\mu}\left(\int \mu q \mathrm{d}s + A\right)}$$ THE solution to the IVP

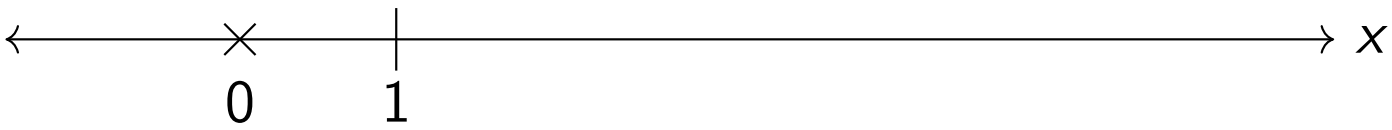
All these integrals can be done where p and q are continuous. A can be determined uniquely from the initial condition.

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Ex: Consider the IVP: $x^2 y' - y = 3;\ \ y(1) = 0$

First rewrite the equation in the form $y' - \frac{1}{x^2} y = \frac{3}{x^2}$

So $p(x) = \frac{1}{x^2}$ and $q(x) = \frac{3}{x^2}$



They both have a single discontinuity at $x = 0$. The IC is given at $x = 1$, so the interval I where a unique solution is guaranteed to exist is $(0, +\infty)$

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Let's find the solution. Use the IF $\mu = \mathrm{e}^{-\int \frac{1}{x^2}\mathrm{d}x} = \mathrm{e}^{\frac{1}{x}}$:

$$y'\mathrm{e}^{\frac{1}{x}} - \frac{1}{x^2}\mathrm{e}^{\frac{1}{x}}y = \frac{3}{x^2}\mathrm{e}^{\frac{1}{x}} \;\rightarrow\; \left(y\mathrm{e}^{\frac{1}{x}}\right)' = \frac{3}{x^2}\mathrm{e}^{\frac{1}{x}} \;\rightarrow\; \mathrm{e}^{\frac{1}{x}}y = -3\mathrm{e}^{\frac{1}{x}} + A$$

$$y(x) = A\,\mathrm{e}^{-\frac{1}{x}} - 3 \qquad \text{(general solution)}$$

Applying the initial condition: $y(1) = A\,\mathrm{e}^{-1} - 3 = 0 \;\rightarrow\; A = 3\,\mathrm{e}$

So **THE** unique solution to the IVP is $\boxed{y(x) = 3\,\mathrm{e}^{(1-\frac{1}{x})} - 3}$

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Theorem of Existence and Uniqueness for 1st Order IVP

Given the 1st order IVP

$$\begin{cases} y' = f(x, y) & \text{(where } f \text{ may be nonlinear)} \\ y(a) = b \end{cases}$$

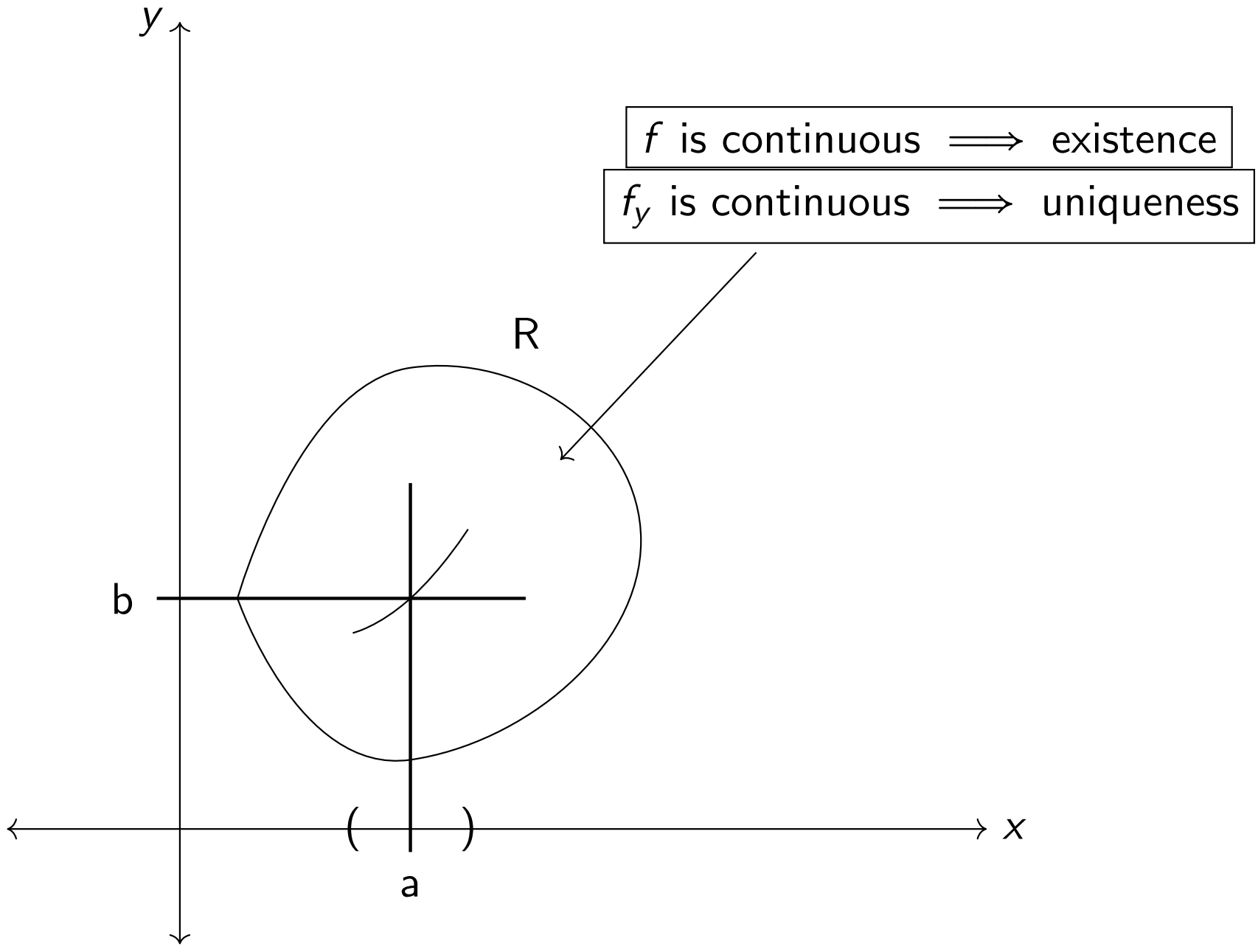
Theorem: If $\exists$ a simply connected region R in the x, y plane that includes (a, b) and if $f(x, y)$ is continuous in R then $\exists$ a solution in an interval around $x = a$. And if $\frac{\partial f(x,y)}{\partial y}$ is also continuous in R then the solution is unique.

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Note: the solution is guaranteed to exist *somewhere* around the initial condition, but the theorem does not say how far it can go (as opposed to the linear case).

Also, if the conditions are not met, a solution may or may not exist, may or may not be unique. It is not an *if and only if* result.

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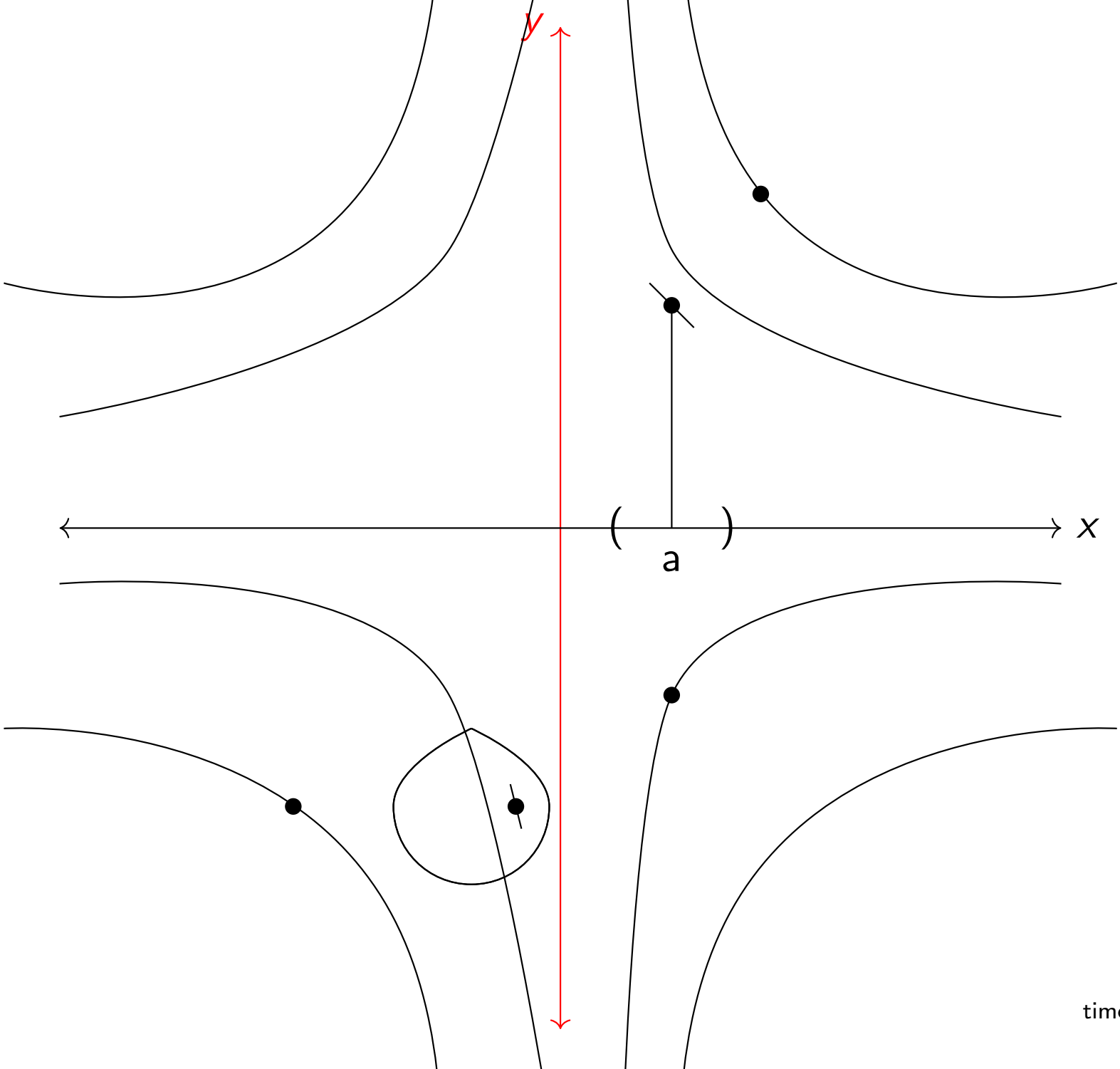
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Ex: $\frac{dy}{dx} = -\frac{y}{x}$

Here $f(x, y) = -\frac{y}{x}$; $\frac{\partial f}{\partial y} = -\frac{1}{x}$

They are both continuous for $x \neq 0$, so if the initial condition is given anywhere but at $x = 0$ the theorem guarantees one and only one solution near the initial condition.

If the initial condition is given at $x = 0$ then the theorem doesn't help.



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Let's solve this separable equation:

$$\frac{\mathrm{d}y}{y} = -\frac{\mathrm{d}x}{x} \;\rightarrow\; \ln(y) = -\ln(x) + A \;\rightarrow\; \boxed{y = \frac{C}{x}}$$

which exists as long as $x \neq 0$ and C can be determined if the IC is not at $x = 0$.

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